

Project Design Phase-II

5	Performance	Access decisions are performed through / server-side scripting
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Technology Stack (Architecture & Stack)

Team ID: SWTID-2026-9038
Project Name: Script-Controlled ACL – Restrict Record Access Based on Field Value

Technical Architecture

Table-1: Components & Technologies

S.No	Component	Description
1	User Interface	User interacts with the ServiceNow application
2	Application Logic-1	Controls record-level READ access based
3	Application Logic-2	Controls CREATE access for users having
4	Application Logic-3	Controls WRITE and DELETE operations u
5	Database	Stores Student Roll Number, Student N Desc
6	Cloud Database	Stores application records within the Servic
7	File Storage	No separate file-storage requirement is spe
8	External API-1	No external API is required for ACL-based
9	External API-2	No external API is required for this project.
10	Machine Learning Model	Machine learning is not required for record
11		/ Cloud) Application and ACL configuration ar

Access Control Design

Operation	Required Role	Condition / Purpo
READ	bb1	Allows access to E
CREATE	bb2	Allows creation of
WRITE	bb3	Allows modificatio
DELETE	bb4	Allows deletion of

Roles Used: bb1 – READ bb2 – CREATE bb3 – WRITE bb4 – DELETE admin – Full access Main Table: u_institution_details

Architecture Flow

User → ServiceNow Web UI → Role & ACL Evaluation → Institution Details Table → Allow / Deny Access

Table-2: Application Characteristics

S.No	Characteristics	Description
1	Open-Source Frameworks	No separate open-source framework is
2	Security	Implements role-based and record-level condition. CREATE
3	Scalable Architecture	Access control is separated by operatio DEL
4	Availability	The application is deployed on the Ser

Fields: Student Roll Number, Student Name, Faculty Name, Branch (ECE/EEE/CSE), Email, Phone Number, Description

Outcome: The project demonstrates script-controlled READ, WRITE, CREATE and DELETE ACLs for record-level security, with user roles and record field values used to determine access.